

# Customer Behavior Analysis

## InternSpark — Data Analyst Task 1

Objective: Analyze customer transactions and behavior to identify customer segments, purchase patterns, and churn risks.

| Metric              | Result                   |
|---------------------|--------------------------|
| Transactions        | 250,000                  |
| Unique customers    | 49,673                   |
| Total revenue       | 681,342,683              |
| Average transaction | 2,725.37                 |
| Churn rate          | 19.95%                   |
| Data period         | 2020-01-01 to 2023-09-15 |
| Duplicate rows      | 0                        |

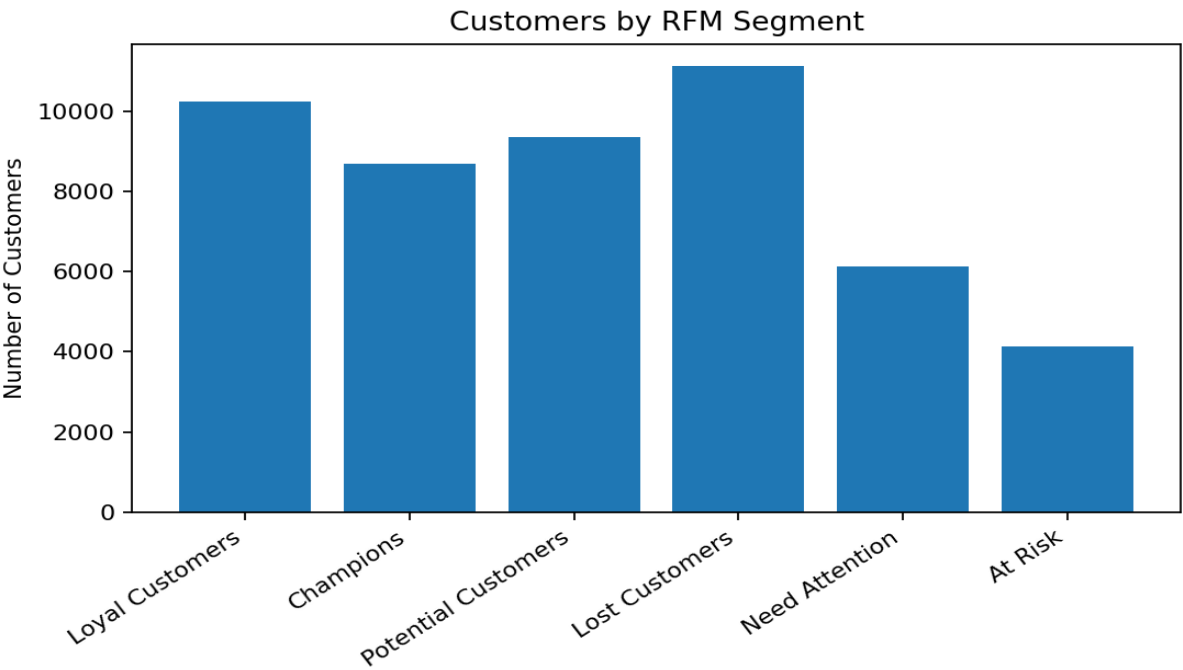
## Data Cleaning

Purchase Date was converted to datetime. The Returns indicator had missing values; for this binary return analysis, missing values were treated as 0 (no return recorded). Duplicate rows were checked and none were found. Month and Revenue features were created for analysis.

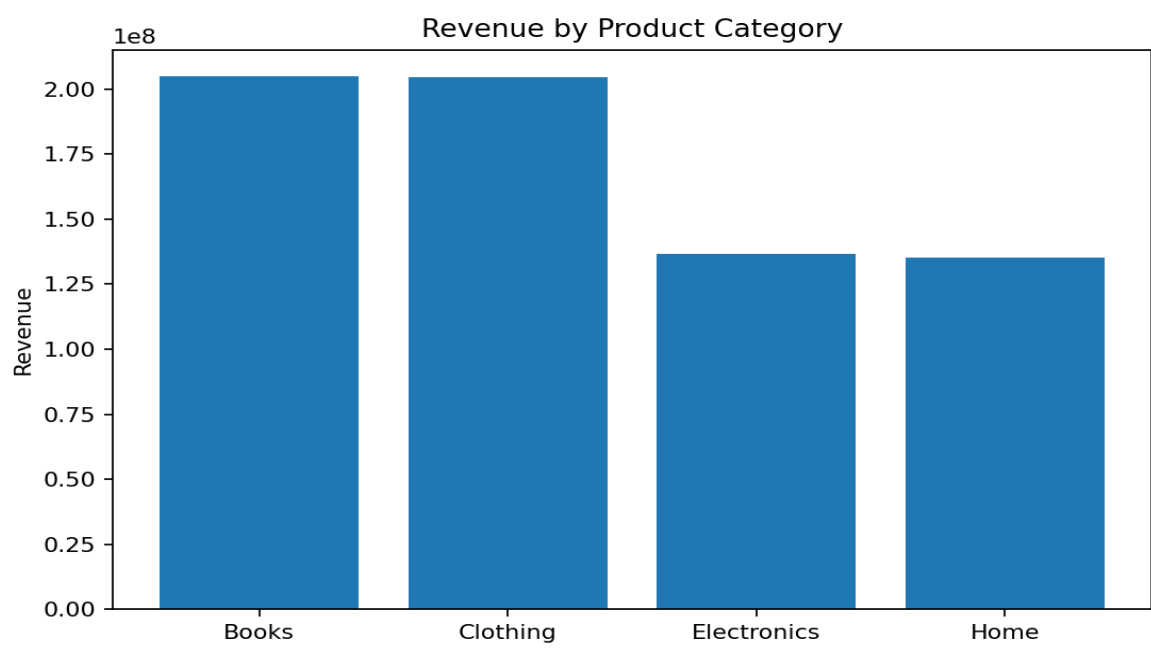
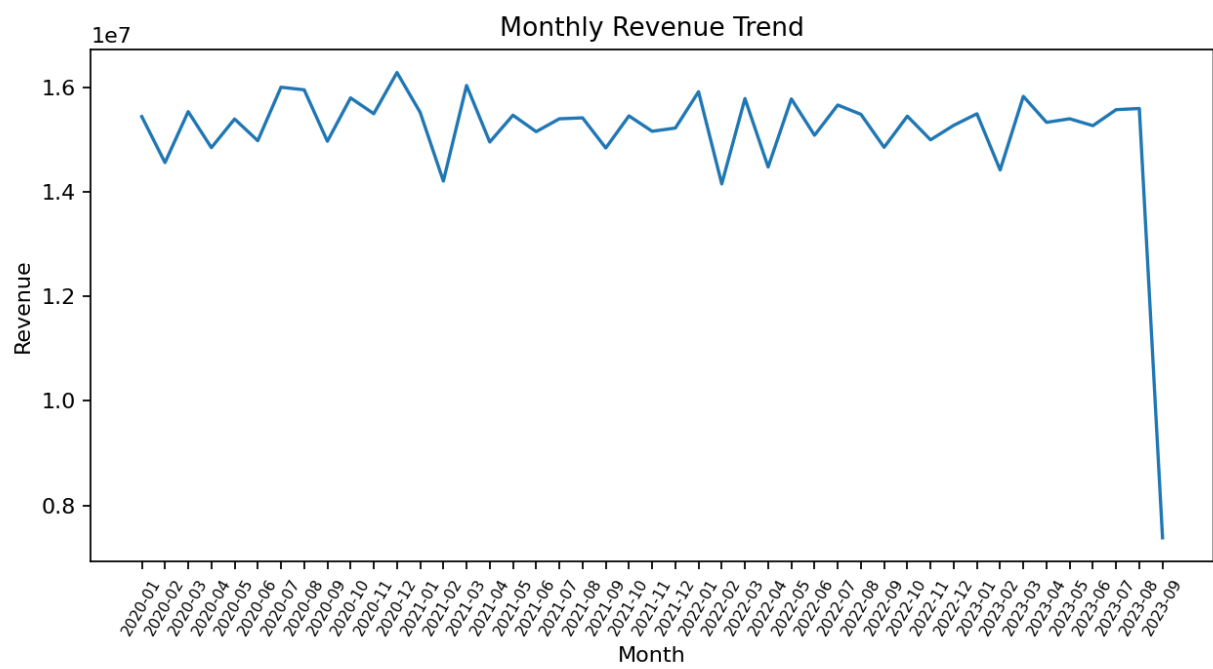
# RFM Segmentation

Recency measures days since the customer's latest purchase, Frequency counts transactions, and Monetary measures total purchase value. Each metric was scored into five quintiles and combined into customer segments.

| Segment             | Customers | Revenue     | Avg Recency | Avg Frequency | Avg Monetary |
|---------------------|-----------|-------------|-------------|---------------|--------------|
| Loyal Customers     | 10,246    | 191,543,261 | 258.3       | 6.9           | 18,694       |
| Champions           | 8,692     | 189,393,382 | 60.9        | 7.6           | 21,789       |
| Potential Customers | 9,348     | 99,780,336  | 66.3        | 3.9           | 10,674       |
| Lost Customers      | 11,125    | 84,336,088  | 564.0       | 2.8           | 7,581        |
| Need Attention      | 6,120     | 63,270,709  | 179.5       | 4.0           | 10,338       |
| At Risk             | 4,142     | 53,018,907  | 438.8       | 4.9           | 12,800       |

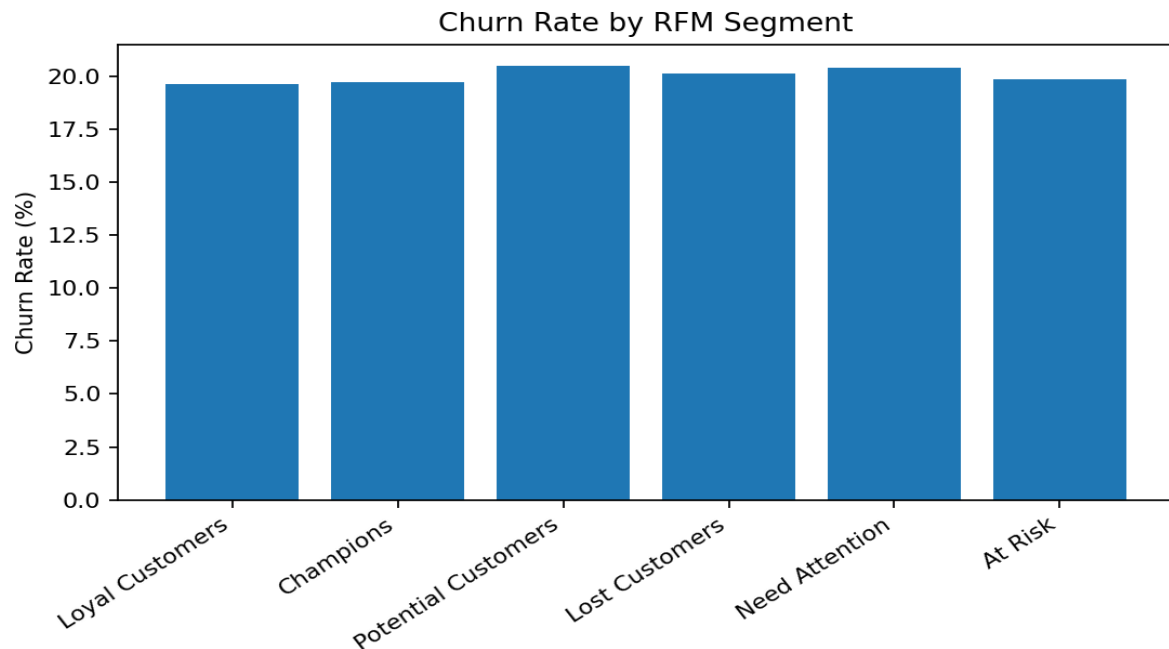


# Purchase Patterns and Retention



## Churn Risk

The dataset contains a churn flag. In this dataset, the overall churn rate is approximately 20%. Because the churn flag is nearly uniform across RFM segments, RFM should be used primarily for prioritization and engagement; churn should be monitored separately rather than assumed to be caused by a segment.



## Five Actionable Recommendations

1. Champions: provide loyalty rewards, early access, and personalized premium offers.
2. Loyal Customers: use loyalty points and relevant product bundles to increase repeat purchases.
3. Potential Customers: trigger targeted second/next-purchase offers and product recommendations.
4. At Risk and Lost Customers: run win-back campaigns with reminders, limited-time incentives, and personalized messaging.
5. Track RFM movement monthly: create a dashboard showing customers moving from Champion/Loyal to At Risk and intervene early.

## Conclusion

The analysis shows a large base of repeat and high-value customers alongside a substantial Lost/At-Risk group. RFM segmentation provides a practical way to prioritize engagement, while monthly purchase and churn monitoring can support retention decisions.